

THE PRODUCTION OF HYDROGEN GAS

Required chemicals:

1. Aluminium foil
2. NaOH
3. Water
4. Match

Required equipment:

1. A glass bottle
2. A balloon

Aim: How to inflate a balloon with hydrogen having water, aluminum and sodium hydroxide

PROCEDURE:

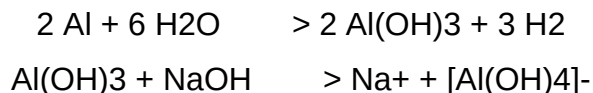
1. Create the water solution of sodium hydroxide
2. Put it into the bottle
3. Add aluminum into the same bottle
4. Put the balloon on the neck of the bottle
5. Wait and control reaction
6. Burn the balloon

DISCUSSION:

You need aluminum, sodium hydroxide and some water. Then, also, a container like a flask, or a bottle (even a plastic bottle would do, but it should be hermetic). Also, the balloon itself. It would be good to have some hygroscopic/absorbent cotton wool (it would be helpful to stop the bubbles not going out of the bottle to the balloon), but it is not necessary if the neck of the bottle is high enough or the reaction speed is controlled. Also, you have to prepare some thread to close the balloon, but it is not necessary if you can close the balloon by making a knot.

Under normal circumstances, aluminum does not react with water, as an impermeable protective layer composed of aluminum hydroxide either forms within seconds or is already in place. With the addition of sodium hydroxide, the

formation of a protective layer is prevented. With the production of aluminates ($\text{Al}(\text{OH})_4^-$), the amphoteric (capable of acting as either an acid or a base) aluminum hydroxide $\text{Al}(\text{OH})_3$ goes in solution:

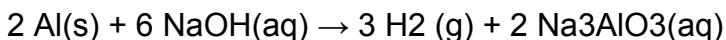


A layer of aluminum oxide previously formed by passive corrosion is dissolved by the addition of sodium hydroxide. For this reason, the reaction takes place at the beginning relatively slowly:



The aluminum completely dissolves and the water acts here too as an acid

CONCLUSION:



Normally this one is one of the ways to make flying balloon but our balloon does not fly .Because the balloon creates some pressure, the actual volume will be smaller.

You don't have to worry much about the amount of water, but it is important to control the concentration of it, because during the reaction, if you add too little of water, it will be much likely that the bubbles (foam-like structure) will develop.